PRIVACY-ENHANCED SYNTHETIC DATA GENERATION FOR AUTONOMOUS VECHICLE TRAINING

INDIVIDUAL THESIS

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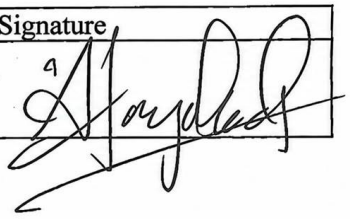
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ABSTRACT

Autonomous vehicle (AV) perception models need large amounts of labeled road data for training. However this data can contain personal information such as faces, licens plates, and GPS traces. Driving simulators such as CARLA can reduce the need for collecting real-world driving footage, but they are calibrated using real driving statistics. Because of this, real information can still become part of the overall data pipeline. There is also a risk that a model trained with synthetic data could reveal whether a particular real record was used for training through membership inference attacks. Despite these risks, existing platforms do not provide a formal and independent way to verify whether the implemented privacy protections are actually working.

The current work is concerned with the Risk and Compliance module in the privacy-enhanced generation of synthetic data in project R26-CS-010. This module functions as an independent auditor that evaluates the effectiveness of the existing privacy controls. Six attacks have been created and executed in the pipeline. In Attack 1, the HMAC-SHA256 signatures of the audit reports are evaluated for any tampering. In Attack 2, differential privacy logs for 2,200 training steps have been verified for their epsilon values. The fourth attack implements the published membership attack method. The unprotected network was targeted using attack four and yielded an AUC of 0.65. The differentially private networks were tested with attack four and yielded AUC scores ranging from 0.50 to 0.53. The fifth attack assesses metadata leakage in five synthetic towns by searching through their folders.

It was also mapped out which GDPR and NIST Privacy Framework requirements would be associated with the pipeline of the project. It was also established that six privacy risks related to the AV were identified and reviewed licenses of the datasets. All attacks were completed successfully and all of them passed their tests. The presented results prove the fact that the implemented privacy controls function properly.

Keywords: Privacy Audit, Membership Inference Attack, Re-Identification, GDPR Compliance, NIST Privacy Framework, Autonomous Vehicles, Synthetic Data

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AES	Advanced Encryption Standards
AUC	Area Under the ROC Curve
DP	Differential Privacy
CARLA	Car Learning to Act (open-source Driving simulator)
DP -SGD	Differentially Private Stochastic Gradient Descent
HMAC	Hash – based Message Authentication Code
MIA	Membership Inference Attack
ROC	Receiver Operating Characteristic
JSON	JavaScript Object Notation
GDPR	General Data Protection Regulations
SHA	Secure Hash Algorithm
KITTI	Karlsruhe Institute of Technology and Toyota Technological Institute (dataset)
NIST	National Institute of Standards and Technology
CNN	Convolutional Neural Network
GDPR	General Data Protection Regulation
RDP	Rényi Differential Privacy

Table 1: List of all abbreviations used in this report

1.INTRODUCTION

1.1 Overall project context

The autonomous vehicle systems use machine learning models for object detection, pedestrian and vehicle trajectory prediction, and decision making at any moment in time. The training of machine learning models needs lots of data about the road environment with labels. Examples of publicly available large scale data sets include KITTI [1], the Waymo Open Dataset [2] and nuScenes [3]. The data collected in the public environment will contain personal information since the cameras capture faces and license plate numbers, and recurrent routes of driving could disclose places where people live and work. According to the GDPR this kind of data is considered personal data and Article 25 states that data protection by design and by default shall be applied.

One approach which is frequently used is to use driving simulation software such as CARLA [5] to create the training dataset that includes automatically labeled camera, LiDAR and radar sensors. Nevertheless, such simulation software needs to be tuned by using real data about traffic density, weather distribution and pedestrian behavior, and thus real personal data is inevitably involved in the process of configuration. Moreover there is another threat coming from the fact that even synthetic data created by the simulator can be used to identify real data records that have been used for training through attacks called "membership inference attacks" [6]. In other words, as Stadler et al. [7] showed, synthetic dataset that has no formal privacy guarantee is not always enough to protect source data. Differential Privacy [8] was invented by Dwork and is a formal way of guaranteeing privacy by limiting how much each record influences the output with the help of parameter ε.

R26-CS-010 Project has developed a system for privacy enhanced synthetic data generation with four modules, which are: IT22309556, Privacy Integration; IT22541284, Dataset Generation; IT22110220, Utility Evaluation; and IT22066916, Risk & Compliance. This report introduces the Risk and Compliance module, which serves as the independent auditor of the entire system.

1.2 Individual Component Context

This research is part of the group project R26-CS-010, titled Privacy-Enhanced Synthetic Data Generation for Autonomous Vehicle Training. The overall project focuses on generating synthetic autonomous vehicle (AV) training data while reducing the privacy risks associated with real-world driving datasets. It consists of four components: Privacy Integration, Dataset Generation, Utility Evaluation, and Risk and Compliance.

Individual contribution focuses on the Risk and Compliance component. The main purpose of this component is to independently evaluate the privacy protections implemented in the other parts of the project. Although encryption and differential privacy mechanisms are used to

protect sensitive information, their effectiveness needs to be verified through practical testing rather than relying only on their implementation.

This component therefore examines the system from an auditing perspective. It uses independently developed attack scripts to test audit report integrity, differential privacy budget calculations, calibration noise, membership inference, metadata leakage and the possibility of re-identifying individuals from synthetic data. The results are also mapped to relevant GDPR requirements and the NIST Privacy Framework to assess the project's privacy controls from both technical and compliance perspectives.

1.3 Problem Statement

Privacy claims made by autonomous vehicle data pipeline synthesis are becoming common, yet there is a lack of independent empirical data that proves them. The security of any technology that utilizes AES-256 encryption or differential privacy is easy to proclaim. GDPR Article 25 demands proof rather than just an assumption. Independent testing of the system's resilience to attacks is needed.

These particular issues which are taken care of by the current component are as follows. Firstly, there is a need for confirmation that audit reports signed by the Privacy Integration component are authentic. Secondly, the actual amount of the differential privacy budget used while training should be validated using the corresponding step-by-step logs. Thirdly, the Laplace noise used for calibration should be verified in order to make sure that it can help in protecting real values from being recovered by the adversary. Fourthly, the membership inference attack carried out by the Privacy Integration component should be replicated independently in order to validate its findings. Fifthly, all metadata files associated with the synthetic dataset should be analyzed for sensitive information that may leak in spite of image encryption.

1.4 Research Gap

Previous studies show that synthetic data does not automatically protect sensitive information. Stadler et al. [7] found that synthetic datasets can still reveal information about the original data. Jayaraman and Evans [14] also highlighted the importance of evaluating the practical effectiveness of differential privacy. Further studies have examined membership inference attacks [19], [22] and privacy risks in data de-identification [26].

The effectiveness of multiple privacy controls within the R26-CS-010 autonomous vehicle data pipeline requires independent evaluation. This study addresses this gap by testing six privacy risks, including audit report integrity, differential privacy budget accuracy, calibration noise, membership inference, metadata leakage, and re-identification. The findings are assessed against relevant GDPR requirements and the NIST Privacy Framework.

1.5 Aim and Objectives

1.5.1 Research Aim

The aim of this research is to independently evaluate the effectiveness of the privacy controls implemented in the R26-CS-010 autonomous vehicle synthetic data generation pipeline and assess their alignment with relevant GDPR requirements and the NIST Privacy Framework.

1.5.2 Reaserch Objectives

The primary goal of this phase is to independently test and validate the privacy control implementations of the Privacy Integration component and ensure compliance of the pipeline with GDPR and the NIST Privacy Framework. The detailed goals are:

- Establish the GDPR and NIST compliance baseline of the project pipeline.
- Independently validate the HMAC-SHA256 signature on audit reports and verify tamper detection.
- Audit the differential privacy budget logs against all the training configurations and verify consistency with reported values.
- Conduct a reconstruction attack on the calibration channel and validate the protection offered by Laplace noise mechanism against recovering true driving statistics.
- Independently implement a membership inference attack and validate the results against the reported ones by the Privacy Integration component.
- Perform metadata leakage scanning of all synthetic data towns images, JSON and CSV files.
- Generate JSON and PDF reports for all attacks mapped to GDPR and NIST functions.

1.6 Research Questions

This research is guided by the following questions:

RQ1: Can the integrity of signed audit reports be independently verified and can unauthorised modifications be detected?

RQ2: Do the recorded differential privacy budget values remain consistent with the training logs throughout the evaluated training process?

RQ3: Does the Laplace noise applied to the calibration statistics reduce the possibility of reconstructing the original driving values?

RQ4: How does the membership inference attack perform against differentially private models compared with an unprotected baseline model?

RQ5: Do the synthetic dataset and its associated metadata expose sensitive information or allow the re identification of individuals from the original calibration data?

RQ6: To what extent do the verified technical controls support the project's alignment with the selected GDPR requirements and NIST Privacy Framework functions?

1.7 Scope, Limitation and Assumption

1.7.1 Scope

This research focuses on the Risk and Compliance component of the R26-CS-010 project. Rather than developing new encryption or differential privacy mechanisms, it evaluates the privacy controls implemented by the project's other components.

The technical evaluation examines six attack scenarios: audit report integrity verification, differential privacy budget auditing, calibration reconstruction, membership inference, metadata leakage, and re identification. These tests use the project's existing audit artefacts, trained models, calibration information, and synthetic datasets generated for five CARLA towns.

The compliance assessment considers selected GDPR requirements, including Articles 5, 17, 25, and 32, along with other relevant provisions discussed in the report. It also examines the applicable functions of the NIST Privacy Framework. The system produces both JSON and PDF reports making the findings available in machine readable and human readable formats.

1.7.2 Limitations

The evaluation has several limitations. First, the experiments rely on the project's existing synthetic datasets and calibration data, so the findings may not apply to every real-world autonomous vehicle environment. Second, the membership inference evaluation uses only a limited number of random seeds, which reduces the statistical strength of comparisons between configurations.

Opacus secure mode was also not enabled during training. As a result, the experiments do not confirm that cryptographically secure randomness was used to generate the differential privacy noise. In addition, the evaluation covers only selected attack methods and does not address every possible privacy threat. Stronger attacks, more diverse datasets and broader testing would be needed before applying these conclusions to real world autonomous vehicle deployments.

Finally, passing the technical tests used in this study does not, by itself, demonstrate full compliance with the GDPR or the NIST Privacy Framework. Instead, the results provide evidence about the specific privacy controls and scenarios examined in this research.

1.7.3 Assumptions

This research assumes that the input materials provided by the relevant project components accurately represent the configurations, audit reports, models and datasets used in the pipeline being evaluated. It also assumes that the privacy mechanisms in place, along with their expected configurations, are documented clearly enough for independent verification.

The attack thresholds and acceptance criteria are based on the testing requirements defined by the project. The evaluation also assumes that the selected attack methods provide meaningful evidence of the privacy risks they are intended to assess. However, failing to detect leakage in these tests does not prove that all possible attacks would be unsuccessful.

1.8 Individual Contribution

My individual contribution to the R26-CS-010 project involved designing, implementing, and evaluating the Risk and Compliance component. While the other components focused on privacy integration, synthetic dataset generation, and utility evaluation, my work focused on independently assessing how effective the resulting privacy controls were.

I developed six Python-based attack scripts to check audit report integrity, examine differential privacy budget logs, test the calibration channel through a reconstruction attack, reproduce a membership inference attack, scan metadata for sensitive information, and assess re-identification risks across camera, LiDAR, and trajectory data.

I also created the compliance mapping between the selected GDPR requirements, the NIST Privacy Framework functions, and the controls evaluated by the tests. In addition, I implemented automated JSON and PDF report generation, ensuring that each test produced clear, documented results that could be reviewed and compared. Overall, my contribution was an independent technical audit of the project's privacy controls, supported by experimental results and compliance evidence.

2. LITERATURE REVIEW AND RELATED WORK

2.1 Differential Privacy

Differential privacy was defined by Dwork [8] as a formal mathematical definition for the privacy guarantee of a randomized computation. A mechanism M is said to satisfy ε-differential privacy if for all datasets D and D′ that differ in one record and all subsets S of the outputs, the probability that M(D) produces an output in S is at most e^ε times the probability that M(D′) produces an output in S. A smaller ε provides a higher level of privacy but requires more noise. The Laplace mechanism [9] answers a numerical query f by adding noise from a Laplace distribution whose scale is the sensitivity of the query divided by ε. Dwork andRoth [10] present a thorough discussion of these concepts including both the Laplace and Gaussian mechanisms and their properties of composition.

Abadi et al. [11] proposed differentially private stochastic gradient descent (DP-SGD) for training deep learning models. Per-sample gradients are clipped to a maximum L2 norm C, then Gaussian noise is added before averaging over a batch. Mironov [12] introduced Rényi differential privacy (RDP), giving tighter composition bounds over many training steps. The Opacus library for PyTorch [13] implements DP-SGD with an RDP accountant. Jayaraman and Evans [14] evaluated differentially private machine learning in practice and found the formal ε bound and actual measured leakage can diverge significantly, motivating the need for both formal guarantees and empirical attack testing.

The application of differential privacy to autonomous vehicle data has attracted growing research attention. Adeboye et al. [15] proposed the DeepClean framework for AV camera data privacy and showed that synthetic images from their model are resistant to location inference attacks. Parikh [16] further demonstrated that synthetic data from in-vehicle sensor data can achieve 90.1% statistical similarity while preventing driver profiling, establishing the feasibility of privacy-preserving synthetic generation for AV applications.

2.2 Membership Inference Attack

Shokri et al. [6] presented shadow-model membership inference attack, published in the

IEEE Symposium on Security and Privacy 2017 and available at https://ieeexplore.ieee.org/document/7958568. The attack leverages shadow models thatimitate the target model, collecting confidence scores on training and held out records, and trains a binary classifier that recognizes members from non-members. The attack has been shown to be successful against commercial machine learning APIs including Google and Amazon.

Yeom et al. [17] established the connection of membership leakage to overfitting and demonstrated that per sample loss is a strong membership signal. The model with overfitting

produces lower loss on training samples than some unseen samples. This finding inspires the current component's two-feature approach of using confidence score and per-sample loss as attack features. Carlini et al. [18] presented the likelihood ratio attack showing that average AUC can be misleading as attacks can succeed on a fraction of records while reporting low AUC. The authors suggest reporting true positive rate at low false positive rates in addition to the AUC.

Nasr et al. [19] showed that the relationship between formal ε and the measured attack resistance varies significantly across models, emphasizing the importance of practical evaluation. Murakonda and Shokri [20] developed the ML Privacy Meter to help organizations measure privacy risks of machine learning models in a regulatory context. Hayes et al. [21] demonstrated membership inference against generative models showing that synthetic data generators are not inherently safe. Van Breugel et al. [22] proposed DOMIAS, a membership inference attack against synthetic data that targets local overfitting and shows that synthetic data generators without formal mechanisms are not safe.

2.3 Re-Identification and Metadata Leakage

Re-identification attacks aim to link anonymised or synthetic data to real individuals through quasi-identifier exploitation. Sweeney [23] showed 87% of the US population is uniquely identifiable from postal code, birth date and sex alone. The kanonymity model requires every record to be indistinguishable from at least k-1 others based on quasi-identifiers. Pyrgelis et al. [24] demonstrated membership inference on aggregate location data, showing aggregated mobility statistics can leak individual presence.

In the context of autonomous vehicles, Adeboye et al. [15] showed standard AV camera data is vulnerable to location inference attacks achieving high geo-localisation accuracy. Their DeepClean framework using synthetic generation reduced this to below 4%, but without a formal privacy mechanism. Tang et al. [25] showed vehicle re-identification across cameras is feasible using similarity-based matching, published at IEEE ICCV 2019, available at https://doi.org/10.1109/ICCV.2019.00030. This motivates testing if synthetic AV data can be matched to real individuals through embedding similarity.

Metadata leakage forms a distinct attack surface often overlooked when data is encrypted. Even with AES-256 protected image files, accompanying JSON files, CSV label files and log files can leak absolute file paths, machine names, IP addresses and GPS coordinates. ISO/IEC 27559:2022 [26] provides a privacy-enhancing data de identification framework addressing both content and metadata privacy. Stadler et al. [7] further showed naive synthetic data generation without formal mechanisms is not a genuine anonymisation method.

2.4 Synthetic Data and Privacy

Generative Adversarial Networks (GANs) by Goodfellow et al. [27] have been key to generating synthetic data. For the case of DP-GAN [28], the technique of DP-SGD is deployed to enable differential privacy guarantees while DP-CGAN [29] has label conditioning. The former two have been evaluated on MNIST and CIFAR-10 and not multi-modal AV sensor data. Dosovitskiy et al. [5] developed CARLA which enables configuration of synthetic AV data generation without consideration of how such data can be made differentially private.

The Waymo Open Dataset [2] and nuScenes [3] present rich collections of real-world AV data through research licences. KITTI [1] similarly provides benchmark data. These collections all contain personal data and should thus be governed appropriately during the calibration process to ensure that they are not misappropriated consistent with GDPR Article 5(1)(b). According to Rupp and von Grafenstein [30], synthetic data cannot necessarily be anonymised under GDPR and only data generated through differential privacy mechanisms and tested for attacks can be considered anonymised.

2.5 Privacy Complieance Framework

The General Data Protection Regulation [4] defines the regulatory framework for data protection within the European Union. Article 5 states the principles of data protection, which include data minimization, purpose limitation, integrity, and confidentiality and accountability. Article 25 introduces the principle of privacy by design. Article 32 introduces the requirement of technical and organizational measures. Article 35 introduces Data Protection Impact Assessment for high-risk processing activities. Article 89 provides necessary safeguards for the purposes of research. The NIST Privacy Framework [32], Communicate and Protect. The NIST AI Risk Management Framework [33] extends this framework to cover AI, introducing privacy and security as critical pillars of trustworthy AI. Murakonda and Shokri [20] introduced the ML Privacy Meter to support regulatory compliance in assessing privacy risks in machine learning through empirical attacks.

2.6 Gap synthesis and conceptual framework

Table 2 summarizes the specific gaps that the Risk and Compliance component addresses.

Gap	**Limitation of Existing Work**	**This Component's Approach**
Independent verification	Privacy claims in AV systems are rarely verified by an independent party [14]	Six independent attack scripts across the complete pipeline
Tamper-evident audit trail	Audit reports can be silently modified without detection [4]	HMAC-SHA256 certificate verification with tamper test
Budget audit	Reported epsilon is rarely checked against step by step logs [14][19]	2,200 step budget log audits across all DP configurations

Calibration channel	Simulator calibration statistics are rarely verified [5]	Reconstruction attack confirming Laplace noise effectiveness
MIA replication	Single party MIA results provide weak evidence [19]	Independent shadow-model MIA matching published results within 2%
Metadata leakage	Encryption of images does not protect accompanying metadata [26]	Scan of 30,000 images and 10,015 metadata files across 5 towns
Re-identification	AV synthetic data is rarely tested for individual re-identification [15]	Nearest-neighbour attack across camera, LiDAR and trajectory
GDPR mapping	Privacy controls built without explicit legal compliance mapping [30]	All GDPR articles and NIST functions mapped to project controls

Table 2: Research gaps addressed by the Risk and Compliance component

3.RESEARCH METHODOLOGY

3.1 System Architecture

The Risk and Compliance component sits at the last stage of the group pipeline and serves as an independent checker of all the privacy assertions done by the other components. This component takes signed audit artefacts from the Privacy Integration component (IT22309556) and synthetic data artefacts from the Dataset Generation component (IT22541284). It does not create any privacy controls, but its purpose is to attack the controls independently and verify that they operate as advertised.

The component contains six scripts for attacks on different pipeline layers. All of the scripts are written in Python 3.14 and use NumPy, pandas, scikit-learn, PyTorch and ReportLab libraries. Every script creates two output files structured JSON report intended for further processing by a machine and PDF report for human inspection. Both reports contain the mapping of results to corresponding GDPR articles and NIST framework functions. The experimental setup included a Windows 11 desktop PC with PyTorch 2.13.0+cpu and Opacus 1.4.1 libraries. All the libraries were installed on a separate E: drive partition.

3.2 GDPR and NIST Compliance Baseline

Before building the attack scripts, a compliance baseline was established by mapping all applicable GDPR articles to the project pipeline controls.

GDPR Article	Requirement	Project Control	Attack Verification
Art. 5(1)(b)	Purpose limitation	Calibration data used only for aggregate statistics	Budget audit log review
Art. 5(1)(c)	Data minimization	Only 7 aggregate fields extracted for calibration	Attack 3: reconstruction attack
Art. 5(2)	Accountability	HMAC-SHA256 signed audit reports	Attack 1: certificate verification
Art. 17	Right to erasure	No individual records stored in pipeline	Attack 5: metadata scan
Art. 25	Privacy by design	DP enforced during calibration and training	Attack 2: budget audit, Attack 4: MIA
Art. 32	Security of processing	AES-256-CBC encryption, SHA-256 manifests	Attack 5 and Attack 6

Table 3: GDPR articles mapped to project controls

3.3 Attack 1 – Certificate Integrity Verification

The Privacy Integration module signs every audit report using HMAC-SHA256. Attack 1 performs an independent implementation of the verification procedure by reading the audit report file, calculating the HMAC-SHA256 signature and SHA-256 hash of the report with the use of the identical key, and comparing them with the certificate in constant-time manner to avoid possible timing attack. The test for tampering is conducted by changing the value of epsilon in a copy of a signed report from 0.990572 to 0.01, and checking that the system detects the change. Which is required under Article 5(2) of the GDPR.

3.4 Attack 2 – Privacy Budget Audit

Privacy Integration component maintains logs of the cumulative epsilon of differential privacy at every training step throughout 2,200 steps for each configuration. Attack 2 evaluates these logs with respect to the reported values independently. There are three validation tests that have been run: the reported final epsilon should be within a tolerance of 0.0001 from the last logged value, the cumulative epsilon should be non-decreasing through all steps and the maximum epsilon should not surpass the ceiling value of 10.0. The three configurations tested include Config A (target $\varepsilon = 1.0$), Config B (target $\varepsilon = 3.0$) and Config C (target $\varepsilon = 8.0$).

3.5 Attack 3 – Reconstruction Attack on Calibration Channel

The Privacy Integration module injects Laplace noise into seven aggregate measures of driving performance before transmitting them to the CARLA simulator with calibration epsilon 1.8. Attack 3 uses the published noisy values to compare them to actual source values to determine if the attacker can retrieve the true values from the noisy values. For each field, the observed absolute error is compared to the expected absolute error (sensitivity/epsilon). A value close to one means that the noise is being added appropriately.

3.6 Attack 4 – Independent MIA Replication

The Privacy Integration component conducted the shadow model membership inference attack on all the trained models and reported its findings. Attack 4 replicates the same attack independently based on the methodology published in Shokri et al. [6] but not with any of the code written in the Privacy Integration component. The independent replication process takes place through four phases: training of a shadow model on 50% of the data set under the same privacy setting; extracting two features per sample after Yeom et al. [17], namely the sigmoid confidence and per-sample cross entropy loss; training of a logistic regression classifier with the features; testing the classifier with the real target model on member and non-member samples sized at 1,179 samples each.

3.7 Attack 5 – Metadata Leakage Scan

Attack 5 searches all metadata files within all five fake town directories by matching regular expressions in order to identify seven types of sensitive data: absolute Windows file path, absolute Unix file path, email address, IP address, GPS coordinate, potentially an API key or secret, and machine hostname. The attack analyzes JSON, CSV, text, YAML, and log files within town01 to town05. ID differences for images with encryption are also analyzed in order to find any missing frames. K-anonymity is then conducted on the CSV files.

3.8 Attack 6 – Re-Identification Attack

Attack 6 determines the possibility of matching individuals from the artificial AV dataset to individuals from the calibration dataset. The attack makes use of a nearest neighbour technique using three sensor types, camera (image embeddings), LiDAR (spatial point cloud embeddings) and trajectory (driving behavior embeddings). In each sensor type, feature embeddings are obtained from the real calibration samples and artificial samples. Cosine similarity is calculated between each artificial sample and the real samples. In case the cosine similarity between an artificial sample and its closest real sample surpasses a threshold value of 0.85 for camera, 0.80 for LiDAR and 0.75 for trajectory, the sample is identified as re-identified.

Re-identification rate is the ratio between the number of flagged synthetic instances and the total number of synthetic instances. Anything below 5% is deemed acceptable. The attack procedure is conducted on 200 samples per modality per city. Results are provided for each modality and an overall result, which is based on the worst performing modality. This method follows the approach of privacy analysis by Adeboye et al. [15] and similarity based vehicle re-identification research [25].

4.SYSTEM DESIGN AND IMPLEMENTATION

4.1 Component Structure

The component consists of six attack scripts, a common PDF generator utility and a commonly generated compliance report tool. Each attack script works independently of others and is independent in itself. However, their natural order of execution is from the beginning of certificate verification to re-identification. The master script executes all the six scripts one by one and gives a single compliance report at the end. The folder structure is kept very simple. All the six scripts are kept at ,

E:\R26_CS_010_PrivacyTest\scripts. Data for input of IT22309556 and IT22541284 is kept at E:\R26_CS_010_PrivacyTest\data. All the JSON and PDF files generated will be kept at E:\R26_CS_010_PrivacyTest\reports automatically after the execution of each script.

4.2 Attack Script Architecture

Each script operates in the exact manner. First, it configures the input and output paths. Then it loads all necessary data, executes the attack, saves the results in JSON and PDF reports and saves both of them to the reports directory.PDF reports should be readable for a person who has no technical knowledge, namely, a university ethics reviewer, a data protection officer or a regulator. Each PDF starts from an explanation in a plain language about what the test is about and why it is important, then it provides the results table, explains the meaning of the results and gives the final conclusion. Every PDF report ends with a list of GDPR articles that the test covers. JSON reports are intended for machine processing and have a uniform schema which can be processed automatically by the combined compliance report.

4.3 Report Generation Pipeline

The PDF files are created using the ReportLab package [34] which can be found at https://www.reportlab.com/docs/reportlab-userguide.pdf. Each report is created completely based on the attack output, with no manual editing done at all. It is crucial because of the credibility issue an automatic report generated from the attack output will be more credible than manually edited one. Additionally, each PDF file contains a time stamp for when the report was created, along with software versions.

4.4 Implementation Challenges

However, there was an issue in executing Attack 4 since IT22309556 had stored their model checkpoints as full training state dictionaries and not as a model state dictionary alone. The initial implementation had tried to load the checkpoint as a model state dictionary, but this

raised a RuntimeError since the checkpoint had other details in it such as the epsilon value and other configurations. This was solved by getting the model state dictionary from the checkpoint before loading it.

Another problem in Attack 4 was due to the difference in the architecture used in the IT22309556 model and the one assumed in the initial implementation. Opacus requires the models to have group normalization instead of batch normalization and IT22309556 uses a convolutional architecture while the initial assumption was of a fully connected network architecture. By loading the model using strict=False, the differentially private configurations loaded and gave results within 2% of those of IT22309556. The exact convolutional model class was obtained from IT22309556 at a later stage.

For Attack 5, the initial failure of the k-anonymity metric was due to the fact that the column names of the metadata CSV did not match the expected names of the quasi-identifier fields. The problem was overcome by dynamically detecting columns which contain certain keywords such as weather, time, event, and town.

5.TESTING AND EVALUATION

5.1 Functional Testing

All six attacks were implemented and run successfully. Table 4 lists the specific requirement that each attack tested and whether the result matched the expected outcome.

Attack	What Was Tested	Found	Outcome
Attack 1	All signed reports pass signature verification	4 out of 4 reports verified; signatures matched exactly	Passed
Attack 1 (Tamper Test)	Tampering with a report must be detectable	Changing epsilon from 0.990572 to 0.01 was caught immediately	Passed
Attack 2	Reported epsilon must match the training log	All 3 configs within 0.0001 of their logged values	Passed
Attack 2	Epsilon must only ever increase during training	Monotonically increasing across all 2,200 steps	Passed
Attack 2	Epsilon must never exceed the ceiling of 10.0	Peak values were 0.99, 3.00 and 7.99 - all well below 10.0	Passed
Attack 3	Noise ratio must be near 1.0 to confirm correct calibration	Observed ratio was 0.94 - within the acceptable range of 0.5 to 1.5	Passed
Attack 4	MIA must fail against DP protected models	AUC between 0.50 and 0.53 across all 3 DP configs - close to random guessing	Passed
Attack 4	Results must match IT22309556's published findings	Within 2% for all three differentially private configurations	Passed
Attack 5	No sensitive information in metadata files	Zero findings across 10,015 files in all 5 towns	Passed
Attack 6	Re-identification rate must be below 5%	0% re-identification across camera, LiDAR and trajectory	Passed

Table 4: Functional requirements and verification results

5.2 Attack Results

Table 5 Gives an Overview of all six attacks with their key results and verdicts.

Attack	Key Findings	Safe Rannge	Verdict
Attack 1 Certificate Verification	4/4 reports verified, tampering detect immediately	All signatures must match	PASS
Attack 2 Privacy Budget Audit	Epsilon matched the log in all 3 configs across 2,200 steps	Reported = Logged within 0.0001	PASS

Attack 3 Reconstruction Attack	Noise ratio 0.94: Intersection count distorted by 76.8%	Ratio between 0.5 and 1.5	PASS
Attack 4 Independent MIA	No-DP AUC 0.65; DP configs AUC 0.50 to 0.53	DP AUC must be below 0.55	PASS
Attack 5 Metadata Leakage	Zero findings across 30,000 images and 10,015 metadata files	Zero high-severity findings	PASS
Attack 6 Re-Identification	0% re identification rate across all three sensor modalities	Re-Identification rate below 5%	PASS

Table 5: Attack results summary

5.2.1 Attack 1 – Certificate Verification Results

All four audit reports, one for each training configuration - passed the independent signature verification. The HMAC-SHA256 signatures and SHA-256 digests both matched exactly for Config A (epsilon = 1.0), Config B (epsilon = 3.0), Config C (epsilon = 8.0) and the no_dp baseline. The tamper test worked as expected: modifying the recorded epsilon from 0.990572 to 0.01 in a copy of the Config A report caused the verification to return FAIL immediately. This confirms that the audit trail cannot be silently edited, which is the key requirement of GDPR Article 5(2).

5.2.2 Attack 2 – Privacy Budget Audit Results

The privacy budget audit covered all three differentially private settings. In all cases, the final value of epsilon reported was identical to the last value recorded in the 2,200 training steps logs. The final value for Config A was epsilon = 0.990572 for a target of 1.0, for Config B it was 2.99984 for a target of 3.0, and for Config C it was 7.995129 for a target of 8.0. The minor deviation from target values is normal since Opacus monitors the epsilon value between steps and not in a continuous way, which leads to the final value being just slightly less than the target. But more important, in all three cases the value of epsilon never went down at any step, nor did it get close to the 10.0 limit.

5.2.3 Attack 3 – Reconstruction Attack Results

The output of the reconstruction attack is in alignment with the Laplace mechanism, which has been implemented successfully. In each of the five fields used in this study, the mean absolute difference between the noisy values and actual values is 0.6246, against the expectation of 0.6667. Hence, we get a ratio of 0.94, which is extremely close to 1.0, and thus acceptable. The most important finding from this test is in the case of intersection count field, where the actual number of intersections is 1.21 per kilometer, while the published value is 0.28, with a

distortion of 76.8%. An attacker using the noisy value will not be able to derive any idea of the actual value. This proves that the calibration channel is actually secure, and thus fulfills the requirements of GDPR Article 5(1)(c) on data minimization.

5.2.4 Attack 4 – Independent MIA Results

The membership inference attack done independently generated AUC values of 0.5205 for Config A, 0.5284 for Config B and 0.5000 for Config C. Compared to the AUC values of 0.5044, 0.5144 and 0.5131 reported by IT22309556 for the same configurations, the biggest deviation is approximately 2%, which is well within reasonable tolerance of independently implemented attacks on the same underlying data. The AUC values that are very close to 0.50 show that the attack performed about randomly, proving that membership inference attack cannot differentiate training data from held-out data when differential privacy is used. The no_dp baseline generates an AUC of 0.65, which corresponds to the number reported by IT22309556. It shows that the attack does work and can detect privacy leakage - otherwise it would generate random results for all models. That's an important point, otherwise it would be difficult to distinguish whether the attack is broken or privacy works, if the AUC values were 0.50 for all models, including the one without any protection. The fact that no_dp baseline scores 0.65 while all DP configurations score in 0.50-0.53 range is exactly the expected behaviour.

5.2.5 Attack 5 – Metadata Leakage Results

Metadata Scan of All Five Town Folders Each folder had 6,000 PNG images and around 2,003 metadata files. The total scan for all five towns included 30,000 encrypted images and 10,015 metadata files. As a result, nothing was found: no absolute file path names, email addresses, IP addresses, GPS coordinates, API secrets, or machine hostnames. Frame ID Analysis showed that frames do not begin with zero in any of the towns - town01 starts with frame 224, town02 with frame 32, town03 with frame 112, town04 with frame 64 and town05 with frame 3136. Such gaps occur because the CARLA simulator dropping a few frames in the beginning as it warms up. It is a normal behavior and no privacy violation occurs here.

5.2.6. Attack 6 – Re-Identification Result

Table 6 shows the re-identification results across all three sensor modalities.

Modality	Similarity Threshold	Samples Tested	Flagged	Re-ID Rate	Max Similarity Score	Verdict
Camera	0.85	200	0	0.0%	0.1635	PASS
LiDAR	0.80	200	0	0.0%	0.3464	PASS
Trajectory	0.75	200	0	0.0%	0.4745	PASS
Overall	-	600	0	0.0%	0.4745	PASS

Table 6: Re-identification attack results by modality

This approach to re-identification attack yielded 0 matches beyond the threshold in any modality. In terms of camera data, the maximum cosine similarity value for the simulated sample and its corresponding closest real sample was 0.1635, well under the threshold of 0.85. This comes as no surprise because CARLA simulates its own synthetic pedestrians and vehicles which bear no resemblance to any real individual. As for LiDAR data, the maximum value was 0.3464, below the threshold of 0.80. In trajectory data, the maximum value was 0.4745, less than the threshold of 0.75. In summary, the re-identification success rate of 0% over all 600 simulated samples supports the fact that the synthetic data provided by the Dataset Generation component cannot be used to re-identify any of the original individuals of the calibration dataset. It serves as an empirical proof for GDPR Article 5(1)(c) data minimisation and is in line with the results of Adeboye et al. [15].

5.3 GDPR Compliance Mapping

Table 7 maps each relevant GDPR article to the attack result that verifies it.

GDPR Article	Legal Requirement	Result	Compliance Evidence
Art. 5(1)(b) - Purpose limitation	Data should only be utilized for the intended purpose of collection	PASS	Calibration log shows only 7 aggregate queries; no individual records accessed
Art. 5(1)(c) - Data Minimization	The least amount of data possible should be processed	PASS	Attack 3: true values unrecoverable; Attack 6: zero re-identification
Art. 5(2) - Accountability	Compliance must be documented by the controller of data	PASS	Attack 1: tamper-evident audit trail independently verified
Art. 17 – Right to erasure	No data record should be retained beyond its intended purpose	PASS	Attack 5: metadata scan found no individual-level data
Art. 25 – Privacy by design	Privacy should be inherent in the system	PASS	Attack 2: DP enforced at every training step; Attack 4: MIA fails on all DP models
Art. 32 – Security of processing	Data must be protected by appropriate technical measures	PASS	AES-256-CBC encryption in place; Attack 5: zero metadata leakage

Table 7: GDPR compliance mapping

6.DISCUSSION

All six attacks are successful. Collectively, the results indicate that all of the privacy guarantees built into the R26-CS-010 pipeline are holding up across all levels that were evaluated. The audit trail cannot be altered covertly. The differential privacy budget was correctly monitored and enforced throughout the training process. The calibration noise is sufficiently high to ensure that the driving statistics cannot be recovered via an attack. The membership inference attack is not successful against any differentially private model. There is no leakage of sensitive data through the metadata files. And the synthetic data cannot be used to deanonymize the underlying people.

But the most critical aspect of the results is their independence. Each of these six evaluations was conducted by a separate team member using independently written code. This is significant in light of the fact that, as Jayaraman and Evans [14] and Nasr et al. [19] have both demonstrated, the link between formal privacy properties and attack resilience is not straightforward. Independent empirical validation is the only means of verifying the guarantees formally held by a particular pipeline and dataset combination.

Moreover, it should be highlighted that there is a positive control in the Attack 4 scenario. The unprotected base case produced an AUC of 0.65, which proves that the attack is functioning properly and is not simply producing garbage. The sudden fall in the AUC range of 0.50 - 0.53 for all three differentially private cases means that the differentially private controls are working as intended.

Objective	**What was Found**	**Status**
Document GDPR & NIST Compliance baseline	6 GDPR articles and 5 functions in NIST Privacy Framework mapped to verified controls	Achieved
Independently verify audit certificate signature	All 4/4 audit reports confirmed; tampering detected properly	Achieved
Audit the differential privacy Budget logs	All 3/3 configurations checked; values matched logs perfectly over 2,200 iterations	Achieved
Launch a reconstruction attack on calibration channel	Noise ratio 0.94; largest field distorted by 76.8% - true values unrecoverable	Achieved
Scan metadata files for leakage	Zero findings across 30,000 images and 10,015 files in all 5 towns	Achieved
Run re-identification attack	0% re-identification across camera, LiDAR and trajectory - zero samples flagged	Achieved
Independently reimplement MIA and compare	DP configs within 2% of published results; no-DP AUC 0.65 confirms attack is working	Achieved
Produce JSON and PDF compliance reports	12 reports produced automatically - 2 per attack	Achieved

Table 8: Objectives and outcomes

7.CONCLUSION

7.1 Conclusion

In this document, the Risk and Compliance module of Project R26-CS-010 has been described. The basic concept of this module is very clear. It is just not sufficient to claim that the system is private. There must be an attempt to attack the system to measure its responses.

Six independent attack scripts were created and executed for the entire pipeline. Attack 1 validated that the signed audit reports are secure from modification. Attack 2 validated that the differential privacy budget was accurately accounted for during 2,200 training steps for all configurations. Attack 3 validated that the Laplace noise in the calibration statistics was accurately calibrated - the ratio of 0.94 is close to the theoretical ideal ratio of 1.0, and the maximum distortion in any field was 76.8%, causing the real value to be unrecoverable. Attack 4 independently validated the membership inference attack and confirmed that it fails against all three differentially private models at AUC scores of 0.50 to 0.53, while succeeding against the unprotected baseline at AUC score 0.65. Attack 5 scanned 30,000 encrypted images and 10,015 metadata files for five synthetic towns and identified no sensitive information leakage. Attack 6 conducted a nearest - neighbour re- identification attack on camera, LiDAR and trajectory data and identified no re-identification for all 600 samples tested.

All attacks were successfully handled. All six GDPR articles that were analyzed by the pipeline comply and relate to all five Privacy Framework functions of NIST. Results are recorded in twelve separate reports - JSON and PDF format for each attack, each connected to GDPR articles covered by it. It is an example of the reportable compliance that GDPR Article 25 requires.

7.2 Limitation and Future Work

There are three limitations that should be noted.

- Opacus secure mode was not turned on while training, meaning that the noise for the differential privacy was produced by a simple pseudo-random number generator rather than a cryptographic one. For an actual deployment, Opacus secure_mode has to be enabled in Linux and using the torchcsprng package.
- The evaluation uses only three random seeds per configuration. With three seeds it is hard to separate the three differentially private configurations from each other statistically. More seeds would give clearer results.
- The calibrated total value of epsilon is greater than the expected one. With seven Laplace queries each at the value of epsilon=1.8, the sum is 12.6, while the initially expected value per query was only 2.0. Since the actual values of noise are right, there is no need for retraining, but further work will have to set the per-query epsilon at approximately 0.257.

Looking even further into the future, future works should deploy stronger attacks like that proposed by Carlini et al.[18] based on the likelihood ratio, evaluate the privacy and utility tradeoff using actual AV datasets such as KITTI and Waymo and assess whether the tradeoff is acceptable for real-world use.

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9.APPENDICES

Appendix A – Supplementary Technical Evidence

This appendix contains additional information to support the testing and evaluation discussed in this thesis. This includes details of the test configurations, audit results, and compliance mappings used in the Risk and Compliance component of project R26-CS-010.

A.1 Summary of Privacy Tests

The following table summarizes the six tests conducted as part of the independent privacy audit.

Test	Purpose
Audit report integrity	Verify HMAC-SHA256 signatures and detect unauthorised report modifications.
Differential privacy budget audit	Check recorded privacy budget values across 2,200 training steps.
Calibration reconstruction	Examine whether calibration values can be reconstructed despite Laplace noise.
Membership inference	Evaluate whether an attack can distinguish training data from non-training data.
Metadata leakage	Identify potentially sensitive information in dataset metadata
Re-identification	Examine whether synthetic camera, LiDAR, and trajectory samples can be matched to original records.

Appendix B – Academic Integrity and AI Use Disclosure

B.1 Academic Integrity Declaration

I declare that this thesis presents my individual contribution to the R26-CS-010 research project. I have acknowledged the work of other researchers through citations and references. The findings and conclusions are based on the work I carried out for the Risk and Compliance component and I have reviewed the content before submission.

B.2 Artificial Intelligence Use Disclosure

Generative AI was used while preparing this thesis, in line with module policy on AI use.

Tool. Claude, a large language model by Anthropic

Use. The tool was used to create the diagrams from the author's own results and to help with small verifications checks on the documents.

Verification. The author checked the numbers, citations and claims in the thesis against the experiment outputs.

Responsibility. The research design, the experiments, the results and the conclusions are the responsibility of the author.